Hamiltonian Monte Carlo with Finite Differences

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Abstract

In this note, I study the feasibility of approximating Hamiltonian Monte Carlo in one dimension, using finite-difference approximations to the gradient of the log-density. By virtue of some fortunate one-dimensional coincidences, and the geometric structure underlying Hamiltonian dynamics, there are indications that this approach may perform much better than one might expect.

1 Basics of Hamiltonian Monte Carlo

Consider the task of sampling from a one-dimensional distribution with density $\pi(x) = \exp(-V(x))$, where V will be referred to as the *potential energy*, or simply *potential*. One approach would be to use *Hamiltonian Monte Carlo* (HMC, [Bet17, Nea11]), a popular instance of the general class of sampling algorithms known as *Markov Chain Monte Carlo* (MCMC).

In HMC, one will in fact generate (approximate) samples from $\mu(x,p) = \pi(x)\nu(p)$, where $\nu(p) = \mathcal{N}(p|0,1)$, and then forgets the values of p, thus obtaining samples from π as desired.

The starting point of HMC is to note that under the flow of *Hamiltonian Dynamics* (HD, [LR04]) for the *Hamiltonian* $\mathcal{H}(x,p) = V(x) + \frac{1}{2}p^2$, that is, the ODE

$$\frac{dx}{dt} = \frac{\partial \mathcal{H}}{\partial p} = p \tag{1}$$

$$\frac{dp}{dt} = -\frac{\partial \mathcal{H}}{\partial x} = -V'(x), \tag{2}$$

the distribution μ is left invariant. This essentially happens because the dynamics preserve $\mathcal{H}$ exactly, and are volume-preserving. The dynamics also possess the lucrative property of being *symplectic*; this is not strictly needed for the preservation of μ, but is useful in other ways. With this in mind, one could iterate Algorithm 1 to generate a Markov chain which leaves μ invariant:

Algorithm 1 Ideal_HMC$(x, p, \Delta T)$

1. At (x,p), replace p by an independent sample from $\mathcal{N}(p|0,1)$.
2. Let (x', p') denote the outcome of running HD for ΔT units of time, and move to (x', p').

Unfortunately, one typically cannot compute the solution of Hamiltonian dynamics exactly, and so must resort to numerical discretisation. A particularly useful discretisation is the *leapfrog* method (see e.g. [HLW03]), sketched here as Algorithm 2:

Algorithm 2 Leapfrog$(x, p, \Delta T, L)$

1. Set $\delta = \frac{\Delta T}{L}$.
2. For $i = 1, \ldots, L$, do
   (a) $p \leftarrow p - \frac{\delta}{2} V'(x)$
   (b) $x \leftarrow x + \delta p$
   (c) $p \leftarrow p - \frac{\delta}{2} V'(x)$.
3. Return (x, p).

This method is *explicit*, which makes it easy to implement, given access to V'. Moreover, it is both *reversible* and *volume-preserving*, which make it easy to incorporate into a Metropolis-Hastings framework. As such, it is a natural candidate with which to build an MCMC algorithm. The most basic such implementation is outlined here as Algorithm 3.

Algorithm 3 Leapfrog_HMC$(x, p, \Delta T, L)$

1. At (x, p), replace p by an independent sample from $\mathcal{N}(p|0, 1)$.
2. Compute $(x', p') = \text{Leapfrog}(x, p, \Delta T, L)$.
3. Compute $\Delta\mathcal{H} = \mathcal{H}(x', p') - \mathcal{H}(x, p)$, and sample $\mathcal{E} \sim \text{Exponential}(1)$.
4. If $\mathcal{E} > \Delta\mathcal{H}$, move to (x', p'). Otherwise, remain at (x, p).

An additional benefit of the leapfrog integrator is that it is *symplectic*, which ensures that even for large values of ΔT, the value of $\Delta\mathcal{H}$ remains bounded, provided $\delta = \frac{\Delta T}{L}$ is sufficiently small. This is particularly useful, as it means that one can propose long moves in space, while retaining a reasonable probability of accepting these moves. The mathematics behind this comes from *geometric numerical integration* [HLW06], and in particular, from the mutually closely-related notions of *backward error analysis* [Rei99], *modified equations* [CMSS94], and of *Shadow Hamiltonians* [IH04].

2 Finite-Difference Hamiltonian Monte Carlo

In some cases, even the leapfrog scheme might not be possible. There are scenarios in which V is not differentiable at all points, and others in which calculating V' is sufficiently expensive that we prefer to think of it as being intractable. As such, we might take some small $h > 0$, and use a (central / symmetric) finite-difference approximation to V':

$$V'_h(x) \triangleq \frac{V(x+h) - V(x-h)}{2h}, \tag{3}$$

and substitute this into our other algorithms wherever V' appears, e.g.

Algorithm 4 FD_Leapfrog$(x, p, \Delta T, L, h)$

1. Set $\delta = \frac{\Delta T}{L}$.
2. For $i = 1, \ldots, L$, do
   (a) $p \leftarrow p - \frac{\delta}{2} V'_h(x)$
   (b) $x \leftarrow x + \delta p$
   (c) $p \leftarrow p - \frac{\delta}{2} V'_h(x)$.
3. Return (x, p).

Algorithm 5 FD_Leapfrog_HMC$(x, p, \Delta T, L, h)$

1. At (x, p), replace p by an independent sample from $\mathcal{N}(p|0, 1)$.
2. Compute $(x', p') = \text{FD_Leapfrog}(x, p, \Delta T, L, h)$.
3. Compute $\Delta\mathcal{H} = \mathcal{H}(x', p') - \mathcal{H}(x, p)$, and sample $\mathcal{E} \sim \text{Exponential}(1)$.
4. If $\mathcal{E} > \Delta\mathcal{H}$, move to (x', p'). Otherwise, remain at (x, p).

A first question concerns whether this is at all sensible to do - are we still getting close to the ideal HMC algorithm, or has our finite-difference approximation thrown it all away? A first glance confirms that at least something has gone right; our integrator is still explicit, reversible, and volume-preserving, so we can at least run it without too much effort, while retaining theoretical guarantees that our samples will asymptotically be drawn from the right invariant measure, π. What is perhaps surprising is that we are doing *even better* than this.

The crux of the matter is that our finite-difference approximation to the gradient of V is the *exact* gradient of an approximation to V. In particular, defining

$$V_h(x) = \frac{1}{2}\int_{-1}^{1} V(x + h\theta)d\theta, \tag{4}$$

one can compute directly that V_h' is the gradient of V_h. Moreover, V_h is close to V; consider

$$\begin{aligned} V_h(x) &= \frac{1}{2}\int_{-1}^{1} V(x + h\theta)d\theta && (5)\\ &= V(x) + \frac{1}{2}\int_{-1}^{1} \left[V(x + h\theta) - V(x)\right] d\theta && (6)\\ &= V(x) + \frac{1}{2}\int_{-1}^{1} \left[V(x + h\theta) - V(x) - h\theta V'(x)\right] d\theta && (7)\\ &\approx V(x) + \frac{1}{2}\int_{-1}^{1} \left[\frac{1}{2}h^2\theta^2 V''(x)\right] d\theta = V(x) + \frac{1}{6}h^2 V''(x) && (8) \end{aligned}$$

to see that $V_h - V$ is of order h^2, uniformly on compact sets (under suitable regularity conditions on V and h), and in particular, along typical trajectories of Hamiltonian dynamics.

A consequence of this observation is that whenever we run FD_Leapfrog_HMC for the potential V, this is is equivalent to running Leapfrog_HMC for the perturbed potential V_h. As such, our numerical scheme is still symplectic, and thus will approximately preserve $\mathcal{H}_h(x, p) = V_h(x) + \frac{1}{2}p^2$ for long times. Moreover, since $V_h \approx V$, one can deduce that the finite-difference discretisation will still lead to bounded $\Delta\mathcal{H}$ over long trajectories, and thus that one can still expect reasonable acceptance rates over long times, at least provided $\frac{\Delta T}{L}$ is sufficiently small.

3 Conclusion

The preceding discussion justifies that using finite-difference approximations in a Hamiltonian Monte Carlo scheme, *in one dimension*, is a surprisingly acceptable approximation, which preserves many of the desirable properties of the ideal HMC method.

It is a worthwhile exercise to justify to oneself that this does not necessarily hold when using standard higher-dimensional versions of finite-difference approximations. In these cases, one typically retains explicitness, reversibility, and volume preservation, and so perhaps not all has been lost.

Another tractable exercise would be to consider using higher-order approximations to V' (e.g. using a 5-point stencil) and deriving the implied potential V_h in this case.

A more ambitious, and perhaps also worthwhile endeavour, would be to explore whether there exist alternatives to finite-difference approximations which, even in higher dimensions, provide tractable gradient approximations, while also interfacing favourably with HMC and its numerical discretisations.

References

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